

Al-Aqsa University

Faculty of Computers and IT

Engineering and Networks Dept.



جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم الهندسة والشبكات

Operating Systems for Intelligent Systems (ENGS2306)

Final Project Assignment

Lecturer: Eng. Mohammed E. Alhams

Semester: Second Semester 2024/2025

Title: Automating the IoT Monitoring Hub Deployment

Weight: This project is the final assessment for the course and carries a significant weight in your final grade.

Objective:

To demonstrate mastery of the course material by writing a single, robust Bash script that fully automates the setup of a secure, functional web-based monitoring hub on a fresh Ubuntu system.

Scenario:

You are a systems engineer tasked with creating a repeatable deployment process for a simple IoT Monitoring Hub. The hub is a web server that displays a status page. Your goal is to eliminate all manual setup steps by creating a single script that handles everything from user creation to firewall configuration.

Detailed Script Requirements (deploy_hub.sh):

Your script must be executable (`chmod +x deploy_hub.sh`) and perform the following tasks in a logical order. It should be run with `sudo` (`sudo ./deploy_hub.sh`).

1. **Error Handling:** The script must start with `set -e` to ensure it exits immediately if any command fails.
2. **User & Group Management:**
 - Create a group named `hub_admins`.
 - Create a user named `iot_admin`.
 - Add the `iot_admin` user to the `hub_admins` group.
3. **Directory & Permissions Setup:**
 - Create a directory at `/srv/iot_hub`.
 - Set the ownership of this directory to `iot_admin:hub_admins`.
 - Set the permissions of this directory to `775`, allowing the owner and group members full control, while others can only read/execute.
4. **Software Installation:**
 - Update the package lists using `apt update`.
 - Install the `nginx` web server without interactive prompts (`apt install nginx -y`).
5. **Firewall Configuration:**
 - Configure `ufw` to deny all incoming traffic by default.
 - Explicitly allow incoming traffic for `ssh` (port 22) and `http` (port 80).

- Enable the firewall non-interactively.

6. Content Deployment:

- Create a basic index.html file inside /srv/iot_hub. This file should contain the following HTML: <h1>IoT Hub Dashboard - System Normal</h1>.
- **Challenge:** You must also configure nginx to serve files from your /srv/iot_hub directory instead of its default location. (Hint: This involves editing the config file at /etc/nginx/sites-available/default using a command like sed).

7. Service Management:

- Enable the nginx service to start on boot.
- Start the nginx service. (Or restart it after changing the configuration).

8. Final Verification:

- The script's final action must be a "health check". Use pgrep nginx inside an if statement to check if the nginx process is running.
- If it is running, print a green "SUCCESS: IoT Hub deployment completed successfully."
- If it is not running, print a red "ERROR: IoT Hub deployment failed."

Deliverables:

1. **The Script (deploy_hub.sh):** Your complete, fully-commented Bash script.
2. **The Report (studentnum-studentname-FinalProject.pdf):** A PDF document containing:
 - The full text of your script.
 - A section-by-section explanation of what your script does and why.
 - A screenshot of your script running successfully from the terminal.
 - A screenshot of the command curl http://localhost running in the terminal and showing your custom HTML message.

Submission: Upload both files to Moodle by the deadline. Standard academic integrity rules apply.